2026 -1 지능형 로보틱스

Minimum Stable Grip Force Control for Deformable Object Manipulation via Tactile-Guided Future Slip Prediction

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2271029 박윤형

Overview

- Background and Objective
- Task Setup and Dataset Collection
- Model Architecture
- Stage1: Future Slip Prediction
- Stage2: Adaptive Control
- Final Results and Analysis
- Conclusion

Background and Objective

01-1 Background

Why Grip Force Matters

Robotic grasping involves a fundamental **trade-off** between **slip** and **damage.**

- **Insufficient** grip force → Object **slip** and task failure
- **Excessive** grip force → Object **deformation or damage**

Too Much Force







→ Robots must maintain the **minimum stable grip force** that prevents slip while avoiding object damage.

01-1 Background

Why Tactile Sensing?

- Slip begins at the **contact interface before** visible object motion.
- Early slip cues are **difficult to observe** with vision alone.
- Tactile sensing captures **contact deformation** and **marker motion**.

Vision

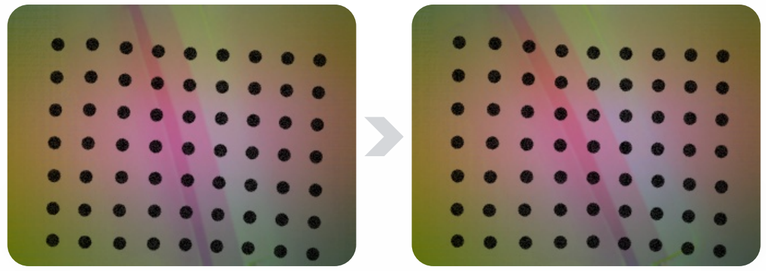
- Object Motion



Tactile

- Contact Deformation
- Marker Motion
- Contact Area Changes

→ Early Slip Cues



01-2 Objective & Pipeline

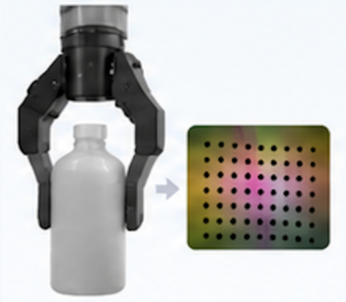
Our Project Goal

Predict slip before it occurs using multi-modal signals and proactively adjust **grip** force to the **minimum stable level**, enabling robots to complete tasks **safely and reliably without causing object damage.**



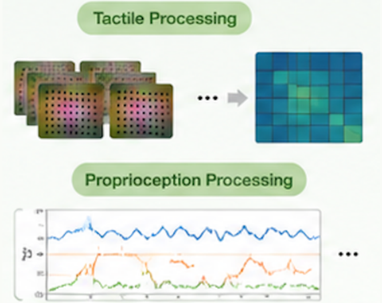
End-to-End Pipeline

1 Data Collection



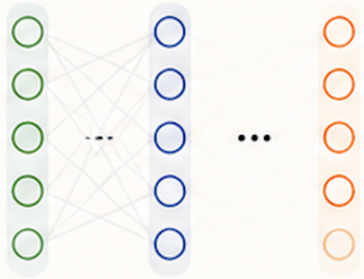
- Simulated grasping
- Multi-modal data: tactile, vision, proprioception

2 Signal Processing



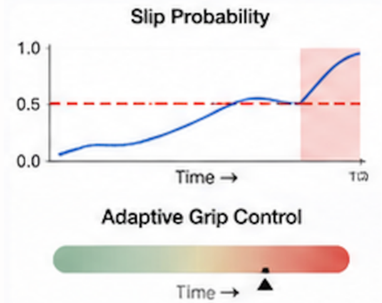
- Tactile processing
- Proprioception encoding

3 Slip Prediction Model



- Learn temporal patterns
- Predict future slip probability

4 Decision & Control



- Early warning
- Adjust grip force proatively

5 Safe Grasping



- Prevent slip
- Avoid object damage
- Complete task successfully

Dataset Collection

02-1 Environment Setup

Environment

- Isaac Sim

Robot

- Franka Emika Panda (7-DoF)

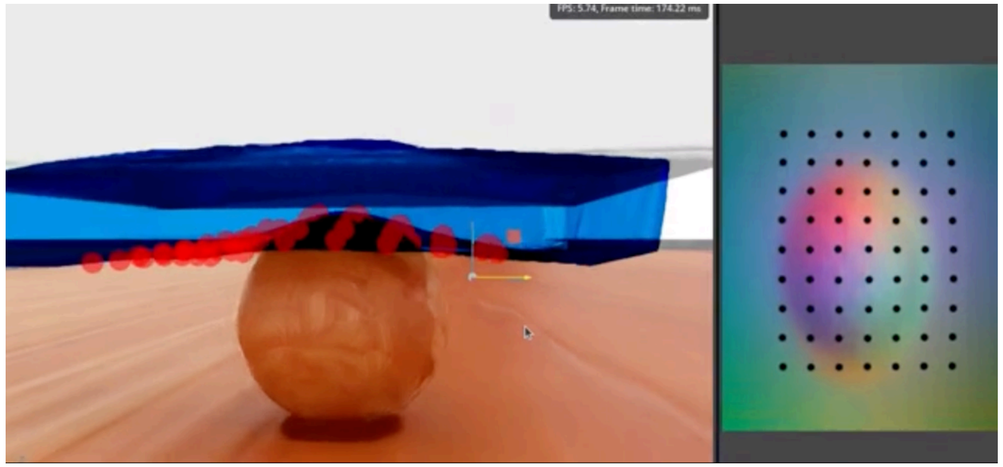
Tactile Sensor

- TacEX (GelSight Mini) x 2 (left/right finger)
- UIPC soft-body contact physics
- Outputs: RGB, depth, marker motion fields

Scene

- Table top manipulation setup
- Wrist camera + head camera





https://sites.google.com/view/tacex

references:

- *UniVTAC: A Unified Simulation Platform for Visuo-Tactile Manipulation Data Generation, Learning, and Benchmarking (https://arxiv.org/pdf/2602.10093)*
- *TacEx: GelSight Tactile Simulation in Isaac Sim–Combining Soft-Body and Visuotactile Simulators (https://arxiv.org/pdf/2411.04776)*

02-2 Task Setup

Objective

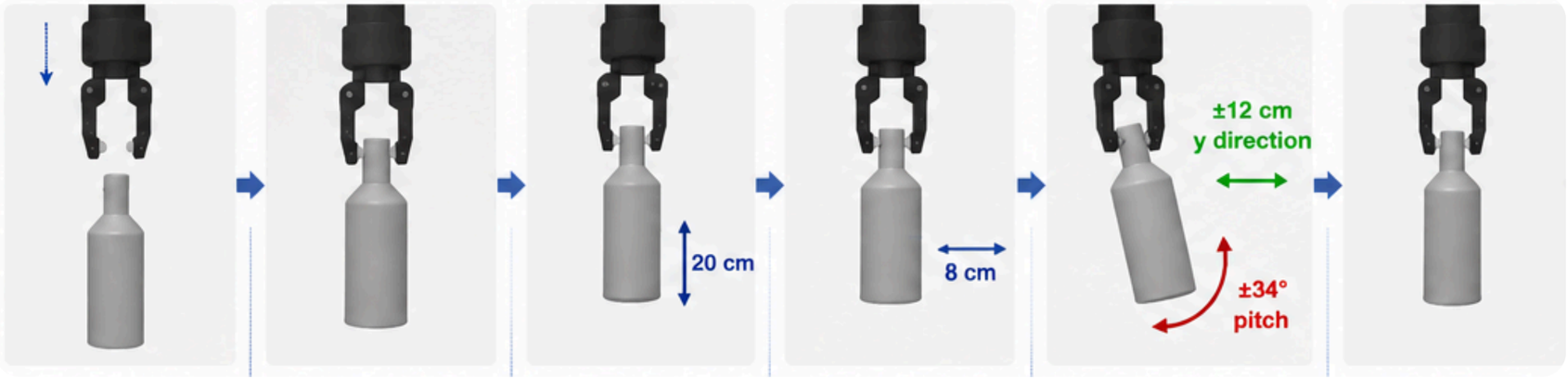
Collect diverse multi-modal data for slip prediction under varying physical conditions on simulation

Objects



Task Pipeline

1 **Approach** 2 **Grasp** 3 **Lift** (+20cm) 4 **Transport** (+8cm, x) 5 **Perturb** (Randomized) 6 **Hold** (20 steps)



Randomized Conditions

Density (UIPC)
100~800 kg/m^3

Grip Force
0.002~0.008 N

Seed
5 seeds

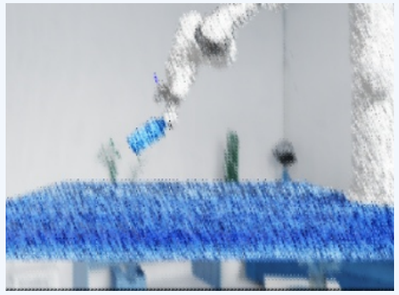


02-3 Data Collection

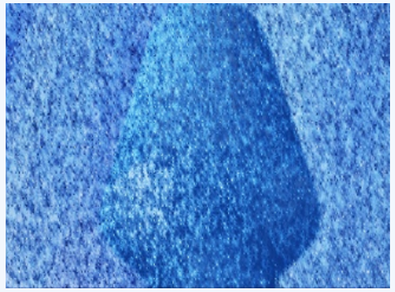
Multi-Modal Observations

Vision

Head RGB



Wrist RGB

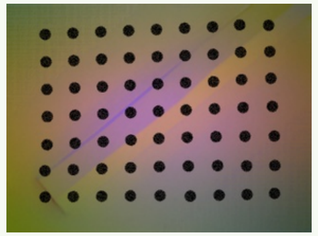


Tactile (Left / Right)

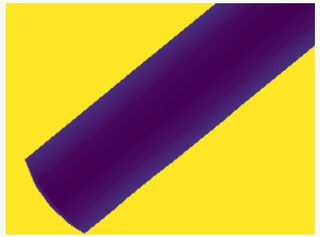
Gelsight RGB



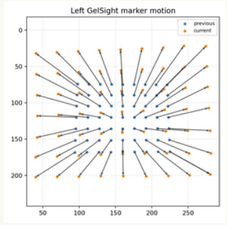
Marker Overlay



Depth

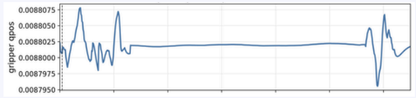


Marker Motion

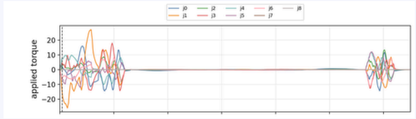


Proprioception

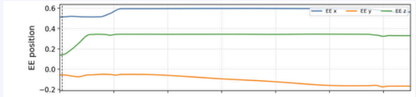
Gripper qpos



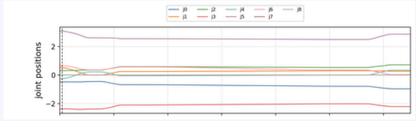
Applied Torque



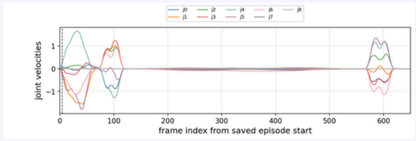
EE Position



Joint Position



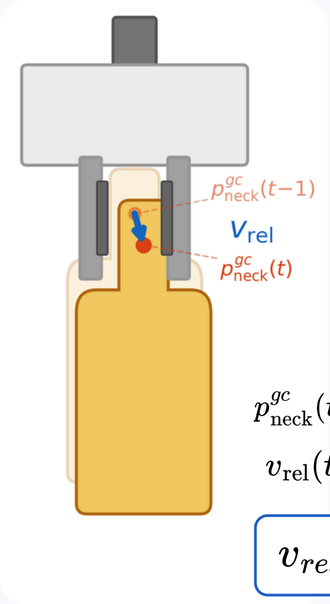
Joint Velocities



02-4 Slip Labeling

Bottle Neck position is tracked in the gripper-center frame for slip detection

Slip Signal (Relative Velocity)



$$p^{gc}_{\text{neck}}(t) = R_{gc}(t)^T \left(p^{\text{world}}_{\text{neck}}(t) - p^{\text{world}}_{gc}(t)\right)$$

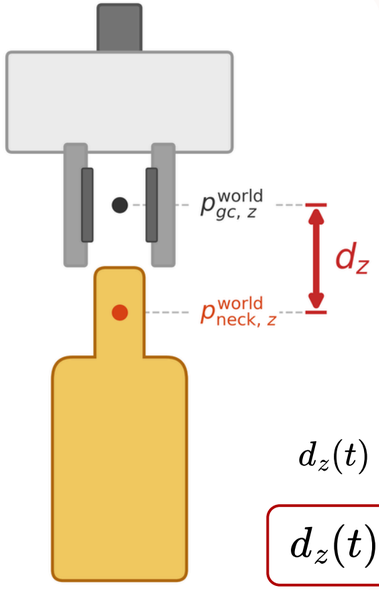
$$v_{\text{rel}}(t) = \left\| p^{gc}_{\text{neck}}(t) - p^{gc}_{\text{neck}}(t-1) \right\|_2$$

$$v_{rel}(t) > 0.03mm \rightarrow \text{slip}$$

$p^{gc}_{neck}(t)$: neck position in gripper-center frame

$R_{gc}(t), p^{world}_{gc}$: gripper pose (rotation, position)

Drop Signal (Vertical Displacement)



$$d_z(t) = p^{\text{world}}_{gc,z}(t) - p^{\text{world}}_{\text{neck},z}(t)$$

$$d_z(t) > 1.5mm \rightarrow \text{drop}$$

$p^{world}_{gc,z}(t)$: gripper center z in world frame

$p^{world}_{neck,z}(t)$: neck contact point z in world frame

Future Slip Label

Slip is detected by the above signals.

If slip occurs within the next H steps (future window) label is assigned as 1.

$$\text{label}(t) = 1$$

if slip occurs in $[t+1, t+H]$

02-4 Slip Labeling - Difficulties

Challanges in Defining Reliable Slip Label Signal

1 Hold Phase Only

Slip Definition

- Object velocity > threshold
- Object pose drift > threshold

Limitation

- Missed slip/drop events occurring during
- lift, transport, and perturb phases

2 Full Task Velocity-Based Detection

Slip Definition

- Object velocity monitored throughout the entire task

Limitation

- Robot motion increased object velocity
- Normal manipulation falsely detected as slip

3 Bottle-Center Relative Velocity

Slip Definition

- Relative motion between bottle center and gripper

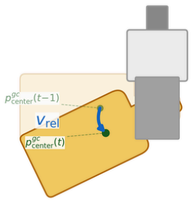
Advantage

- Independent of robot arm movement

Limitation

- Difficult to define a single threshold

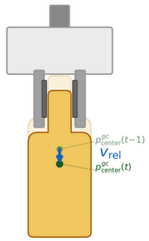
Rotation slip



large v_{rel}
(center sweeps arc)

easily detected

Vertical-only slip



small v_{rel}
(center barely moves)

hard to detect

4 Final Method: Neck Point Tracking

Slip Definition

- Track bottle neck position in the Gripper Center Frame

Advantage

- Measures only contact-point motion
- Detects subtle pre-slip behavior

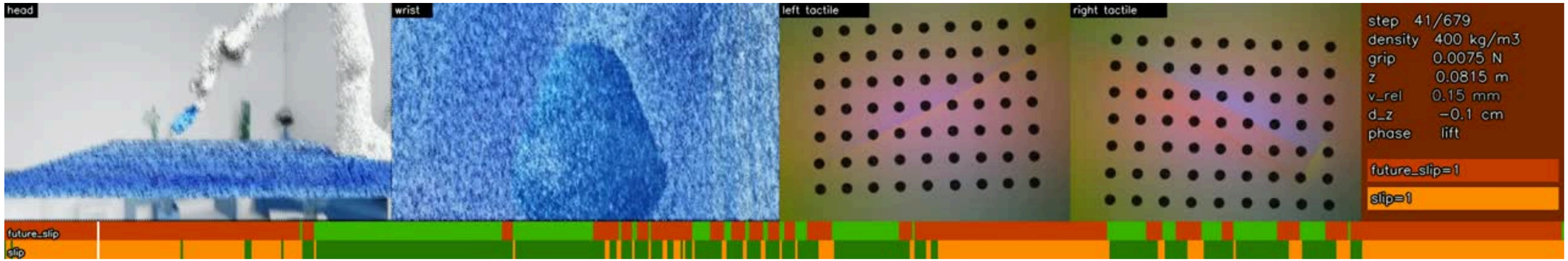
Threshold Selection

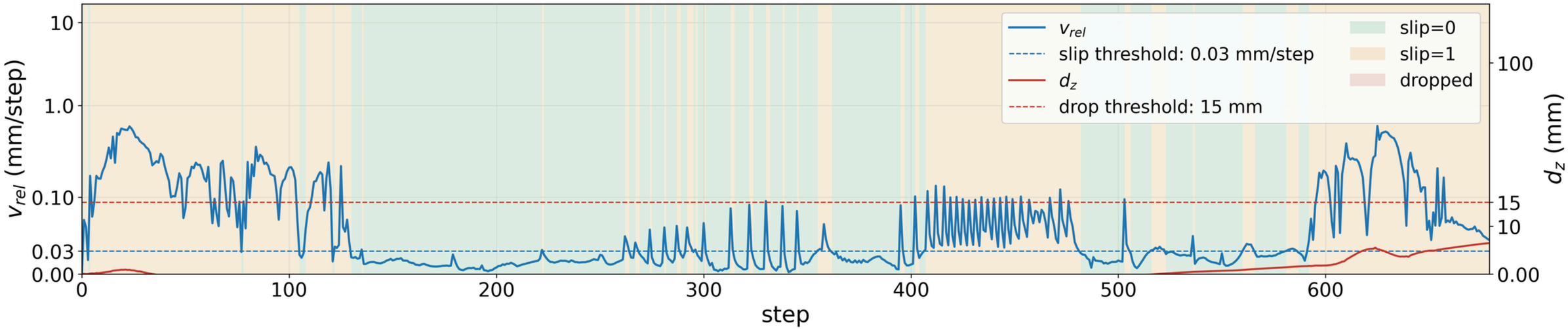
Validated checking tactile RGB observations and relative motion manually

✅ Slip Threshold: **0.03** mm/step

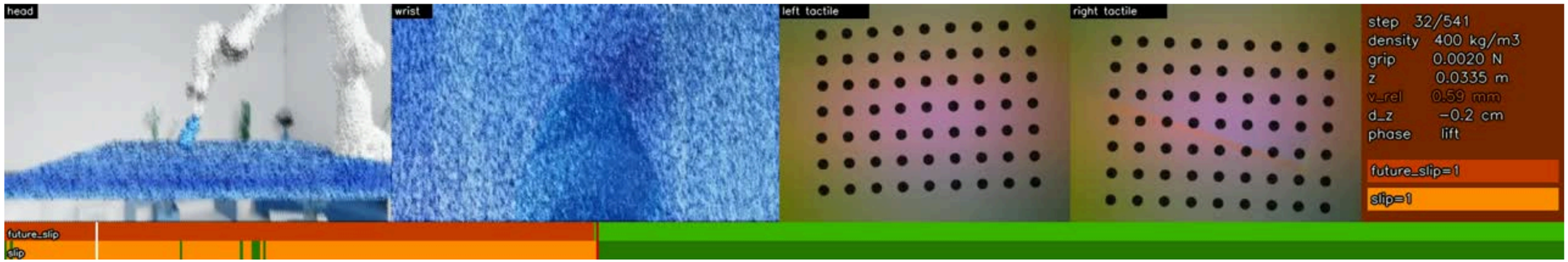
✅ Drop Threshold: **1.5** mm

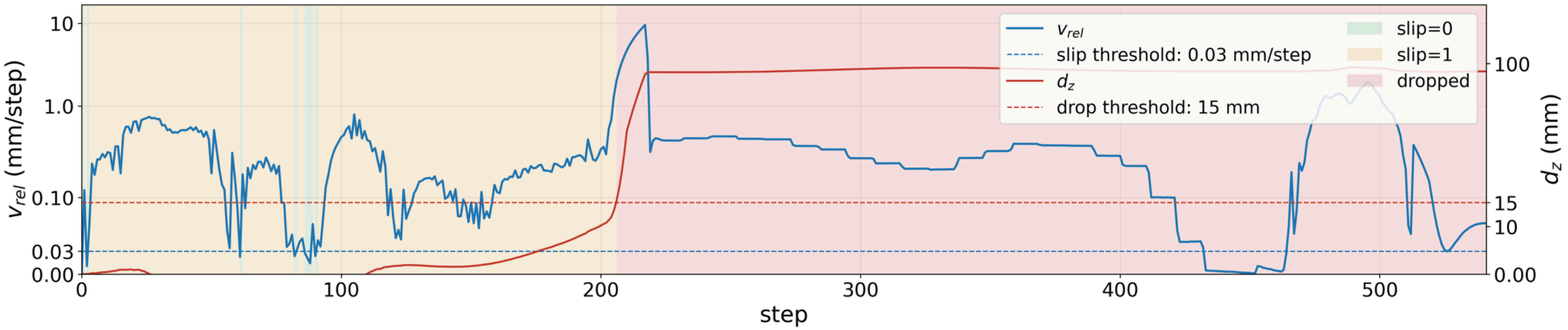
02-4 Slip Labeling





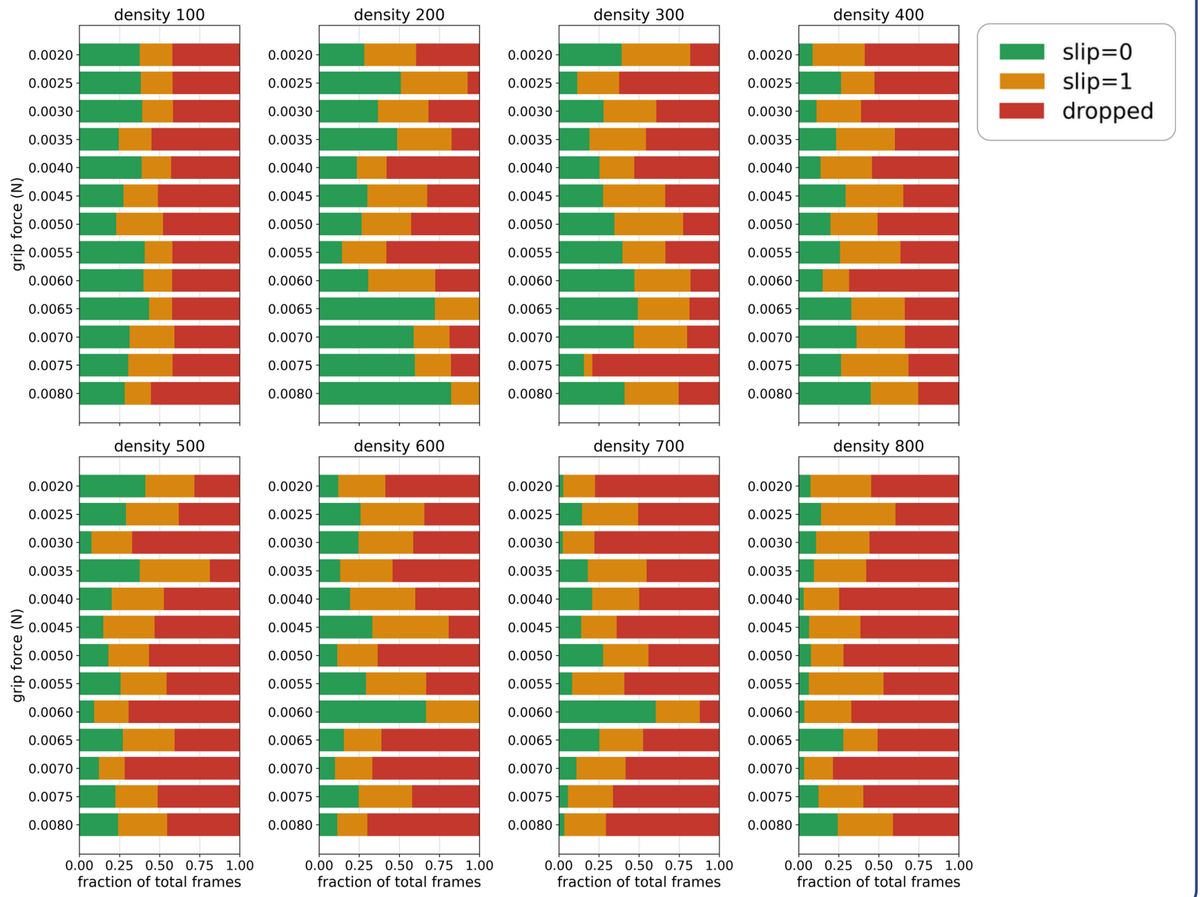
02-4 Slip Labeling



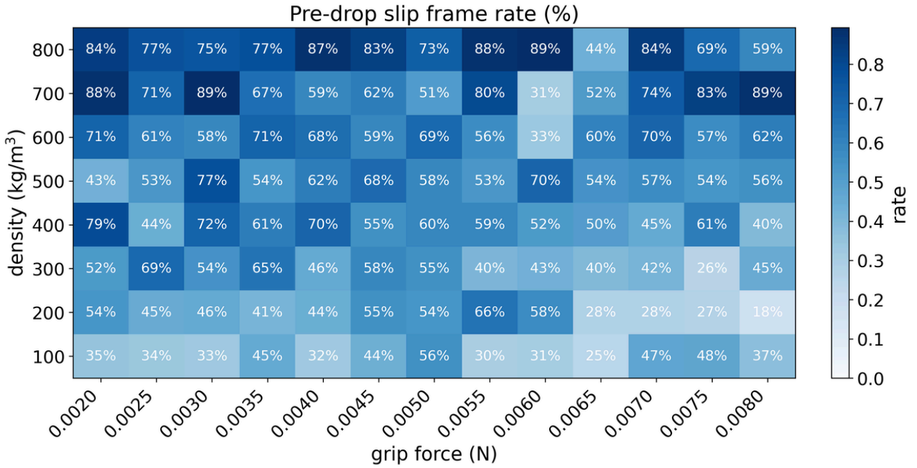


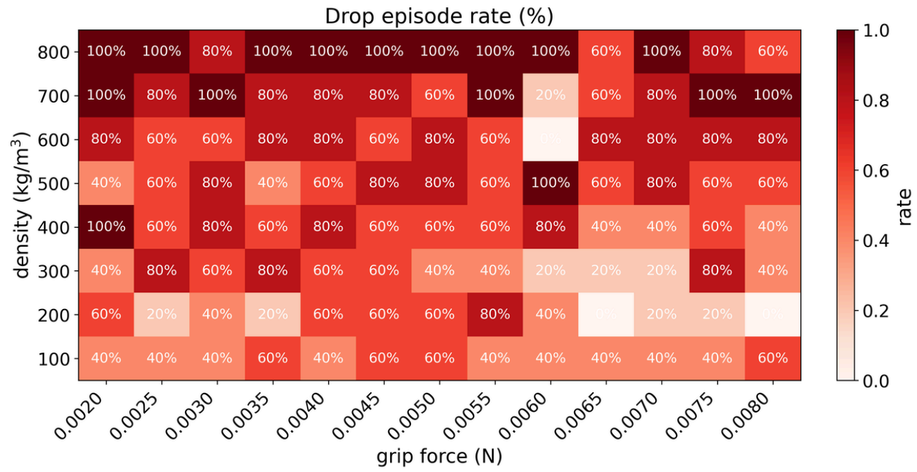
02-5 Dataset Statistics

Distribution Overview



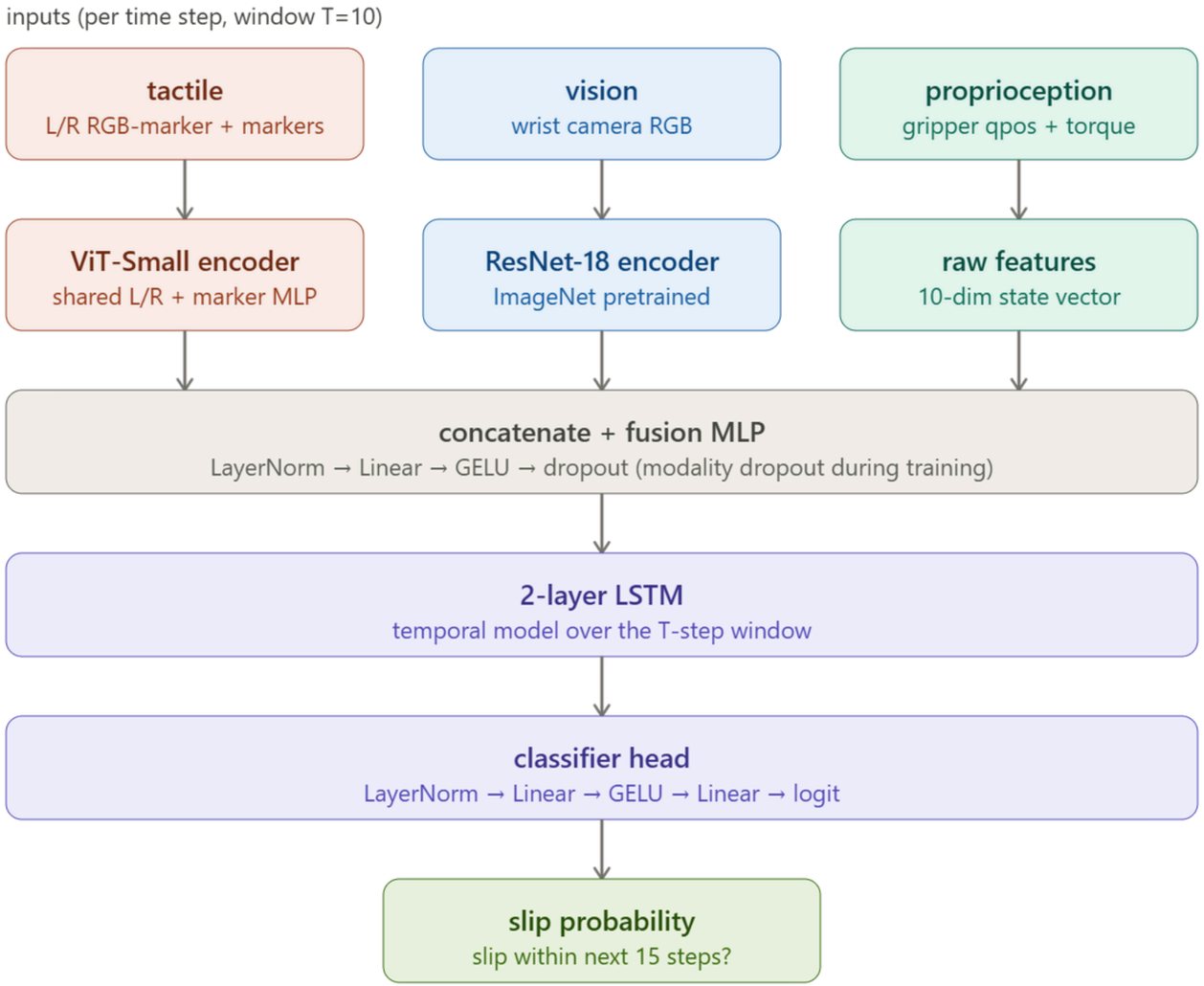
✓ **High slip/drop ratio at high density & low grip force**





Model Architecture

03 Model Architecture



Stage 1
Future Slip Prediction

04-1 Training Implementation

1.Training Configuration

- Batch Size: 32
- Training Epochs: 20
- Early Stopping: Based on validation AUC (patience = 4)

2. Regularization & Overfitting Control

- Weight Decay: 0.001
- Dropout: 0.5
- Modality Dropout: 0.3 (forces use of both modalities)

3. Input & Data Settings

- Modalities: Tactile, Vision, Proprioception
- Image Size: 224
- Future Window: 15

4. Evaluation & Data Split

- Target Metric: AUC
- Split Mode: Stratified split (Fixed per label tag)

04-2 Experiment Results: Occlusion Test

Setup: fixed threshold 0.35 for all three models / epoch 2

Condition	Scenario	Corruption	Severity
Clean	Healthy camera (baseline)	None	—
Blackout	Camera fully blocked	All pixels → 0	100%
Noise	Sensor noise / blur	+ Gaussian noise	std = 0.5
Patch	Object/dirt on lens	Center square → 0	50% area

Table 1. Vision occlusion conditions used in the robustness test. Each condition simulates a different camera failure mode applied to the wrist camera at test time.

모델	CLEAN	BLACKOUT	NOISE	PATCH	최대 하락
vision	0.934	0.793	0.830	0.821	−0.141 ⬇
tactile	0.944	0.944	0.944	0.944	0.000 ✓
full	0.953	0.949	0.938	0.944	−0.015

T**able 2.** AUC of each modality under the four camera conditions (0.03 labels, balanced 50:50, threshold 0.35).

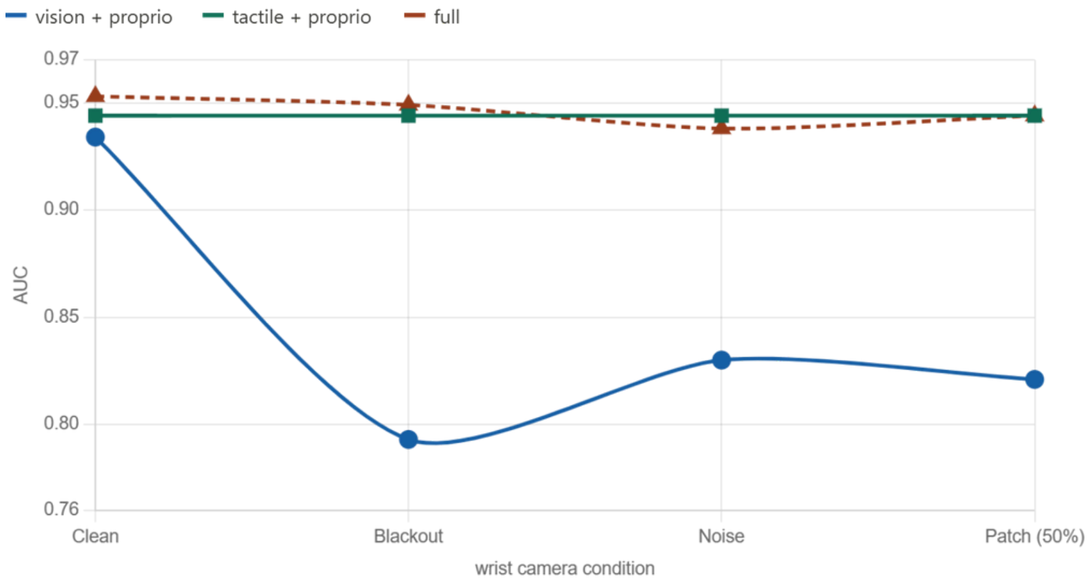


Figure 1. AUC vs. camera condition for the three modalities.

04-2 Experiment Results: Occlusion Test

Setup: fixed threshold 0.35 for all three models / epoch 2
Key findings:

- On clean input, all three models perform comparably (AUC ≈ 0.95).
- Tactile + proprio is completely immune to camera corruption (AUC stays 0.952 across all conditions) — it uses no vision input.
- Vision + proprio degrades sharply under blackout and patch (AUC 0.96 → 0.85).
- Full holds up well (AUC ≥ 0.94 under blackout/patch) because tactile compensates for corrupted vision.

Takeaway:

- Tactile provides a reliable fallback when vision fails — multimodal fusion stays robust under sensor corruption.

0.03 occlusion robustness (balanced 50:50, threshold 0.35):

상황	모델	AUC	f1	recall	precision
CLEAN	vision	0.934	0.856	0.888	0.826
	tactile	0.944	0.845	0.935	0.771
	full	0.953	0.881	0.912	0.852
BLACKOUT	vision	0.793	0.647	0.479	0.996
	tactile	0.944	0.845	0.935	0.771
	full	0.949	0.863	0.903	0.826
NOISE	vision	0.830	0.636	0.468	0.990
	tactile	0.944	0.845	0.935	0.771
	full	0.938	0.686	0.990	0.526
PATCH	vision	0.821	0.607	0.436	0.999
	tactile	0.944	0.845	0.935	0.771
	full	0.944	0.868	0.881	0.856

Table 3. Occlusion robustness: AUC / F1 / recall / precision under wrist-camera corruption (fixed threshold 0.35, epoch 2).

04-2 Experiment Results: Occlusion Test

Setup: fixed threshold 0.55 for all three models / epoch 2

Threshold ablation (0.03 labels):
AUC is threshold-independent (identical at both);
f1/recall/precision shift only marginally (<0.01) with the same trend across all models and conditions.

We report 0.35 as the main operating point.

모델/상황	AUC (둘 다)	f1 @0.35	f1 @0.55	차이
vision CLEAN	0.934	0.856	0.858	+0.002
tactile CLEAN	0.944	0.845	0.852	+0.007
full CLEAN	0.953	0.881	0.883	+0.002
vision BLACKOUT	0.793	0.647	0.645	−0.002

Occlusion robustness — threshold 0.55 ablation (0.03 labels, balanced 50:50):

상황	모델	AUC	f1	recall	precision
CLEAN	vision	0.934	0.858	0.884	0.834
	tactile	0.944	0.852	0.919	0.793
	full	0.953	0.883	0.905	0.863
BLACKOUT	vision	0.793	0.645	0.477	0.997
	tactile	0.944	0.852	0.919	0.793
	full	0.949	0.864	0.899	0.832
NOISE	vision	0.830	0.630	0.462	0.992
	tactile	0.944	0.852	0.919	0.793
	full	0.938	0.691	0.987	0.531
PATCH	vision	0.821	0.604	0.433	0.999
	tactile	0.944	0.852	0.919	0.793
	full	0.944	0.868	0.874	0.862

Table 4. Occlusion robustness: AUC / F1 / recall / precision under wrist-camera corruption (fixed threshold 0.55, epoch 2).

04-3 Unseen-object generalization

Setup: Models trained only on the bottle (0.03 labels), evaluated on a can without retraining.
Both test sets balanced 50:50, fixed threshold 0.35.

Key findings:

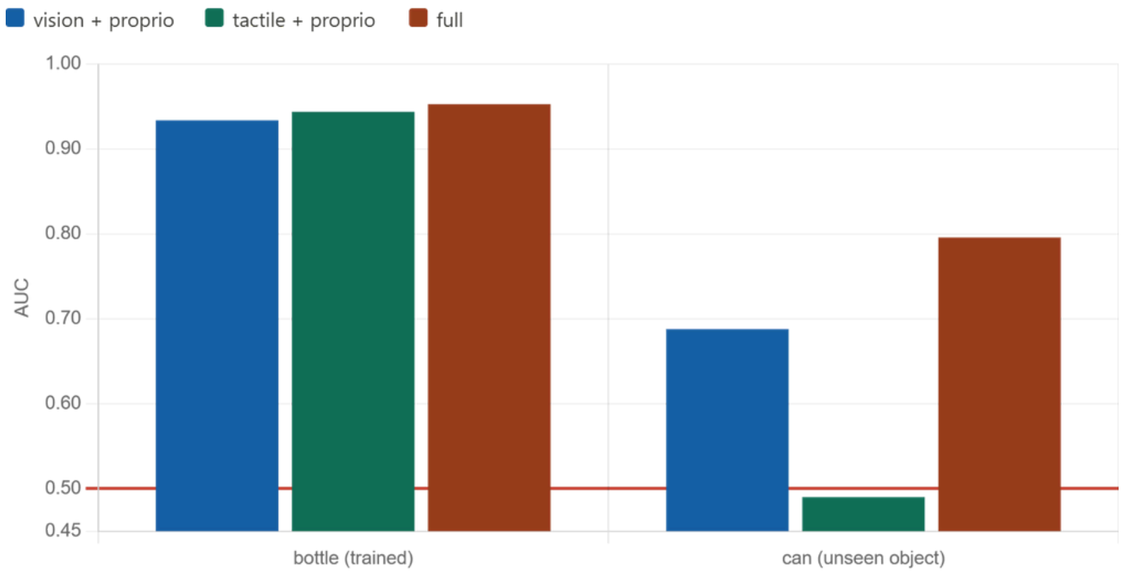
- On the trained object (bottle), all three modalities perform well (AUC 0.93–0.95), including tactile.
- On the unseen object (can), tactile collapses to near-random (AUC 0.49), predicting "always slip" (recall 1.0, accuracy 0.50).
- Full degrades the least (AUC 0.80) and ranks best on the unseen object; vision drops to 0.69.

Takeaway: Under the loose 0.03 labels, the tactile model overfits to the trained object's contact patterns and loses all discrimination on a new shape. Label definition critically shapes out-of-distribution behavior.

Model	AUC	F1	Recall	Precision
Vision + Proprio	0.688	0.625	0.591	0.664
Tactile + Proprio	0.490	0.667	1.000	0.500
Full	**0.796**	**0.708**	0.884	0.590

Model	Bottle (trained)	Can (unseen)	Drop
Vision + Proprio	0.934	0.688	−0.246
Tactile + Proprio	0.944	0.490	**−0.454**
Full	0.953	0.796	−0.157

Table 5. Unseen-object generalization: AUC on the trained object (bottle) vs. an unseen object (can).



04-3 Unseen-object generalization

Setup: 0.03 labels, evaluated on HDMI (unseen) without retraining. Balanced 50:50, threshold 0.35.

Key findings:

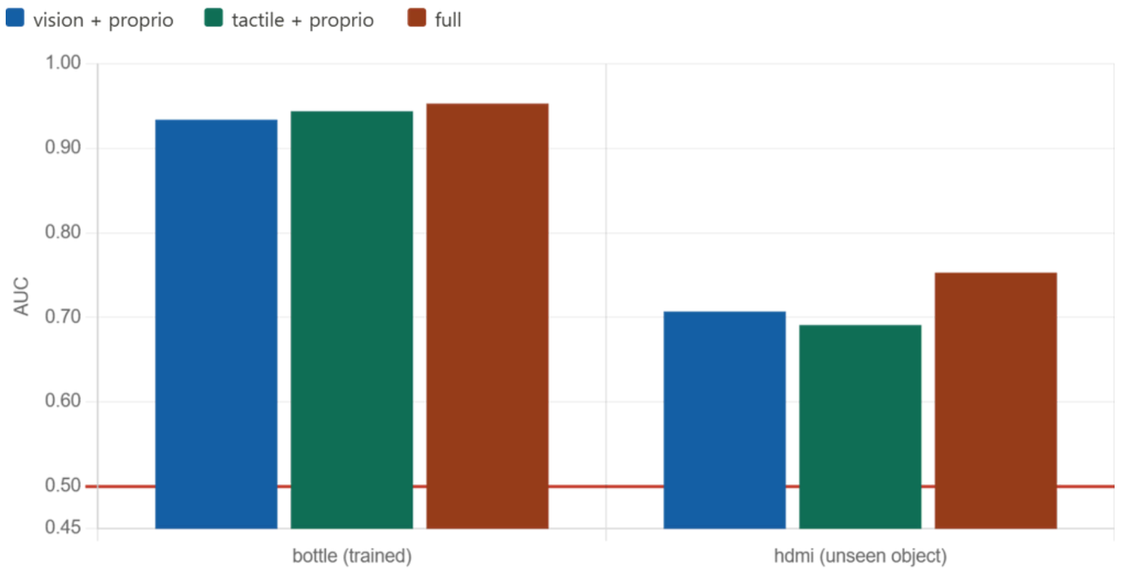
- On hdmi, all three models drop to AUC 0.69–0.75; generalization to a new object is hard for every modality.
- Full ranks best (0.753), as it did on the can — combining tactile and vision gives the most stable transfer.
- Tactile alone varies by object (can 0.49, hdmi 0.69), showing single-modality transfer is less reliable under 0.03 labels.

Takeaway: Under loose 0.03 labels, multimodal fusion (full) generalizes most reliably to unseen objects; no single modality is consistently best.

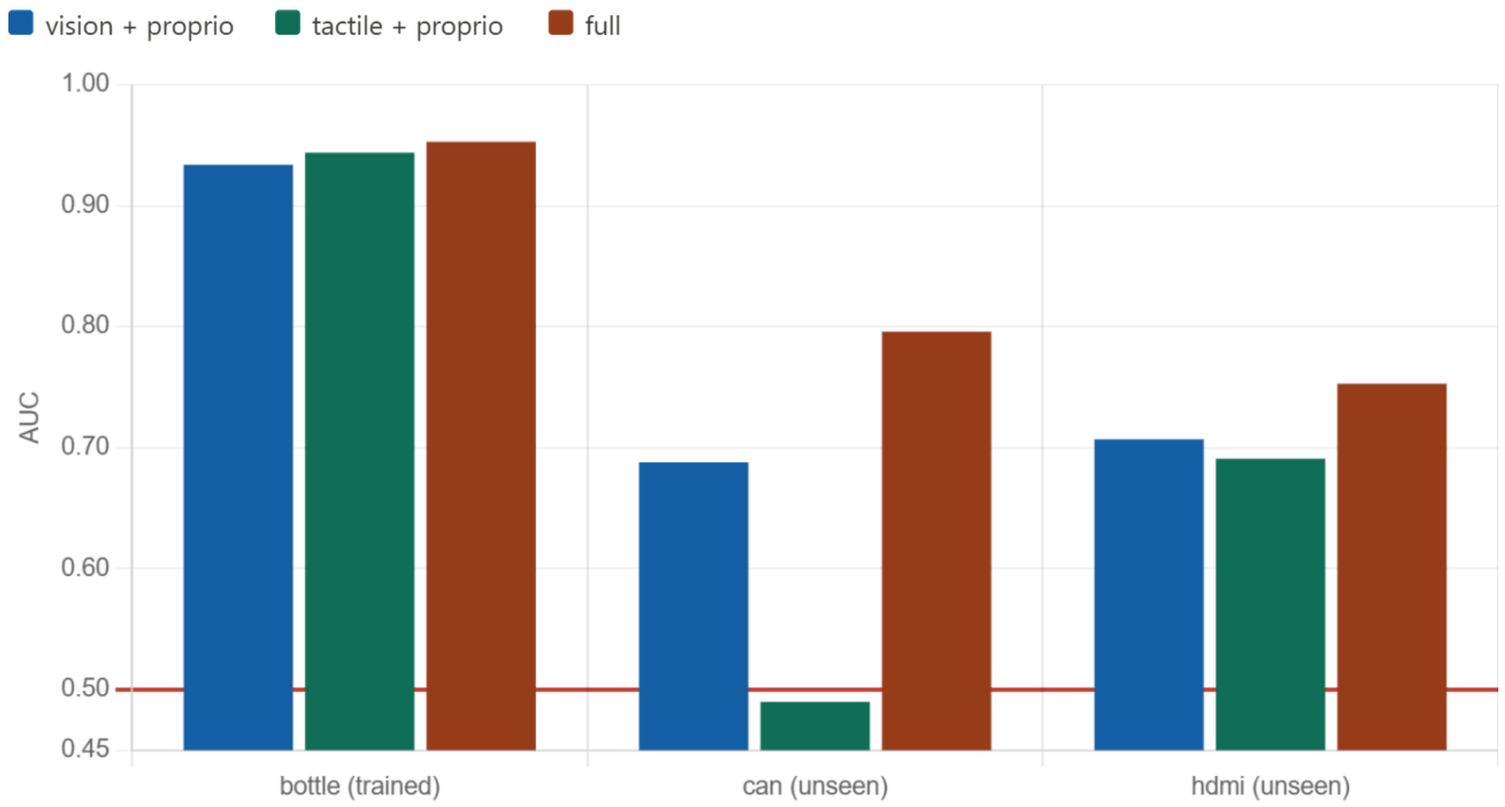
Model	AUC	F1	Recall	Precision
Vision + Proprio	0.707	0.662	0.985	0.499
Tactile + Proprio	0.691	0.665	0.912	0.524
Full	**0.753**	0.671	0.987	0.508

Model	Bottle (trained)	HDMI (unseen)	Drop
Vision + Proprio	0.934	0.707	−0.227
Tactile + Proprio	0.944	0.691	−0.253
Full	0.953	0.753	−0.200

Table 6. Unseen-object generalization: AUC on the trained object (bottle) vs. an unseen object (HDMI).



04-3 Unseen-object generalization



Red line = 0.5 (random). 0.03 labels, balanced 50:50, threshold 0.35.

Stage 2
Adaptive Force Control

05-1 Adaptive Force Control

Overview

- Stage 1's SlipPredictor outputs P(slip): probability of slip within the next 15 steps
- Stage 2 uses P(slip) as a real-time control signal to actively adjust grip force during holding
- Closed-loop pipeline: predict → act → re-observe

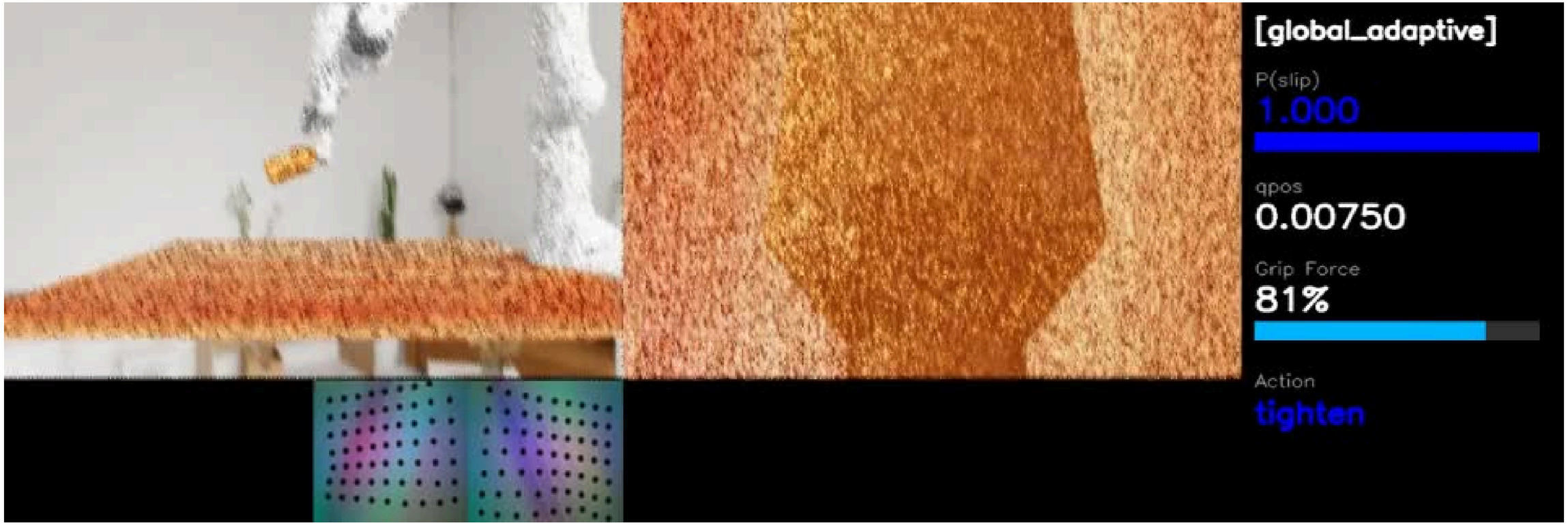
System pipeline

- Each step: collect GelSight tactile + wrist camera + gripper proprioception
- Stack a 10-step window → SlipPredictor → P(slip)
- Controller selects an action from P(slip):

P(slip)	Action	Effect
> 0.35	**Tighten**	increase grip force (↓ qpos)
0.15 – 0.35	**Hold**	maintain force (dead zone, avoids oscillation)
< 0.15	**Relax**	slowly reduce force (seek minimal grip)

- Threshold 0.35 = Stage 1's learned slip/no-slip decision boundary
- qpos ↔ grip force are inversely related (lower qpos = tighter grip)

05-2 Demo


[global_adaptive]
P(slip)
1.000
qpos
0.00750
Grip Force
81%
Action
tighten

Conclusion

06 Conclusion

Significance

- Demonstrates slip probability from multi-modal sensing (tactile + vision + proprioception) can drive real-time control
- Proactive, not reactive: adjusts force before slip occurs, rather than detecting it afterward
- A complete closed-loop manipulation pipeline built on the learned predictor

THANK YOU